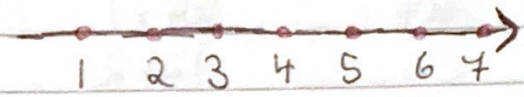
24/09/20

1st Years : Revision

- $\mathbb{N}$ = natural numbers = 1, 2, 3, 4, 5, 6....

positive whole



Q. Is this a natural number?
- 3
- $4\frac{1}{2}$
- -2
- 101

- Even : 2, 4 ... can be divided by 2
  Odd : 1, 3 ...

Q. List the first ten odd numbers
List the first ten even natural numbers

- Ascending : from smallest to biggest
  Descending : from biggest to smallest

Q. Re-arrange these digits: 4, 3, 1, 7

Largest 4-digit number : Smallest:

Q. Write these numbers in order: 3297, 2988, 3079, 3190, 931, 309

Ascending : 931, 2988, 3079, 3098, 3190, 3297
Descending : 3297, 3190, 3098, 3079, 2988, 931

- A factor divides evenly into a number, with no remainder.

Q Factors of 24:

24×1
12×2
8×3
6×4

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

(i) Factors of 18:

18×1
9×2
6×3

Factors of of 18: 1, 2, 3, 6, 9, 18

Highest Common Factor: 6.

Factors of 12:

1×12
2×6
3×4

Factors of 12: 1, 2, 3, 4, 6, 12

(ii) Factors of 24:

1×24
2×12
3×8
4×6

1, 2, 3, 4, 6, 8, 12, 24

HCF: 12

Factors of 36:

1×36
2×18
3×12
4×9

1, 2, 3, 4, 9, 12, 18

- Prime number only has two factors, itself and 1.

Are these prime numbers?

Q. 2 3 4 5

* List first 8 prime numbers:

2, 3, 5, 7, 11, 13, 17, 19

- Prime factors.

12

2×3

$2 \mid 12$
$2 \mid 6$
$3 \mid 3$
1

$2 \times 2 \times 3$

24

$2 \mid 24$
$2 \mid 12$
$2 \mid 6$
$3 \mid 3$
1

$2 \times 2 \times 2 \times 3$

30

$2 \mid 30$
$3 \mid 15$
$5 \mid 5$
1

$2 \times 3 \times 5$

- Squares and Square Roots

Q. $2 \times 2 \times 2 = 2^3$ $\quad 7 \times 7 \times 7 \times 7 \times 7 = 7^3$

$2 \times 2 = 2^2$

$3 \times 3 \times 3 \times 3 = 3^4$ $\quad 8^4 = 8 \times 8 \times 8 \times 8$

$\sqrt{9}$ = 3 $\quad \sqrt{25}$ = 5 $\quad \sqrt{64}$ = 8 $\quad \sqrt{49}$ = 7

BIMDAS

Brackets
Indices (powers)
Multplication
Division
Addition
Subtraction.

Q. $\sqrt{100} + \sqrt{144}$

$10 + 12$

$22.$

$\sqrt{81} + 3^2$

$9 + 9$

18

$\cancel{\frac{3^2 + \sqrt{81}}{3^3}}$

$\cancel{\frac{9 + 9}{27} = \frac{18}{27}}$

$\frac{2^2 + \sqrt{36}}{\sqrt{25}}$

$4 \times (5+6)$

4×11

44

$48 - 3 \times (9-3)$

$48 - 3 \times 6$

$48 - 18$

30

$\frac{\sqrt{81} + 5}{4 \times 4 - 2}$

$\frac{9 + 5}{16 - 2}$

$\frac{14}{14}$

Array Method for Multiplication.

22 x 33

~~765 x 18~~

	~~10~~	~~8~~
~~700~~	~~7000~~	~~56~~
~~60~~		
~~+ 5~~		

426 x 11

	10	+ 1
400	4000	400
+ 20	200	20
+ 6	60	6

4000
400
200
60
20
6
4686